

Hojas de entrega para escanear y subir al Campus Virtual.

Las **hojas de entrega previa** deben ser impresas, rellenadas con sus soluciones, escaneadas o fotografiadas, y subidas al Campus Virtual en la tarea correspondiente.

DNI :

32898935-B

SUM(DNI) mod 7 :

5

Parte 1. Diseño teórico y creación de los esquemáticos.

Tabla de Verdad Decodificador BCD a 7 Segmentos:

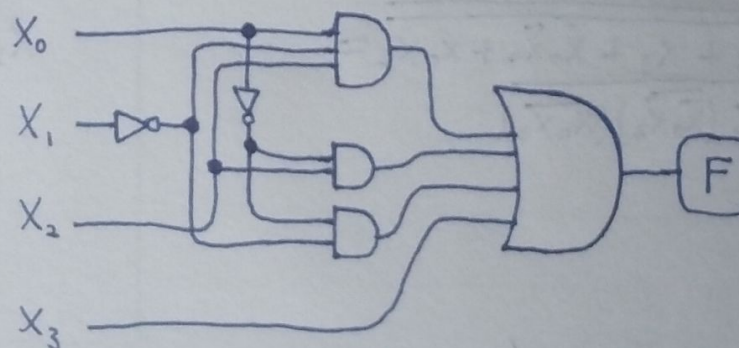
X ₃	X ₂	X ₁	X ₀	A	B	C	D	E	F	G	DP
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	6	0
0	0	1	0	1	1	0	1	1	0	1	0
0	0	1	1	1	1	1	1	0	0	1	0
0	1	0	0	0	1	1	0	0	1	1	0
0	1	0	1	1	0	1	1	0	1	1	0
0	1	1	0	1	0	1	1	1	1	1	0
0	1	1	1	1	1	1	0	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1	0
1	0	0	1	1	1	1	1	0	1	1	0
1	0	1	0	X	X	X	X	X	X	X	0
1	0	1	1	X	X	X	X	X	X	X	0
1	1	0	0	X	X	X	X	X	X	X	0
1	1	0	1	X	X	X	X	X	X	X	0
1	1	1	0	X	X	X	X	X	X	X	0
1	1	1	1	X	X	X	X	X	X	X	0

SM (Suma Mínima)

Segmento F.....

X ₃ , X ₂	00	01	11	10
X ₁ , X ₀				
00	1	1	X	1
01	0	1	X	1
11	0	0	X	X
10	0	1	X	X

Diagrama del Circuito



Función Lógica

$$F = X_0 \bar{X}_1 X_2 + \bar{X}_0 X_2 + \bar{X}_0 \bar{X}_1 + X_3$$

PM (Producto Mínimo) Segmento G

X_3, X_2	00	01	11	10
X_1, X_0				
00	0	1	X	1
01	0	1	X	1
11	1	0	X	X
10	1	1	X	X

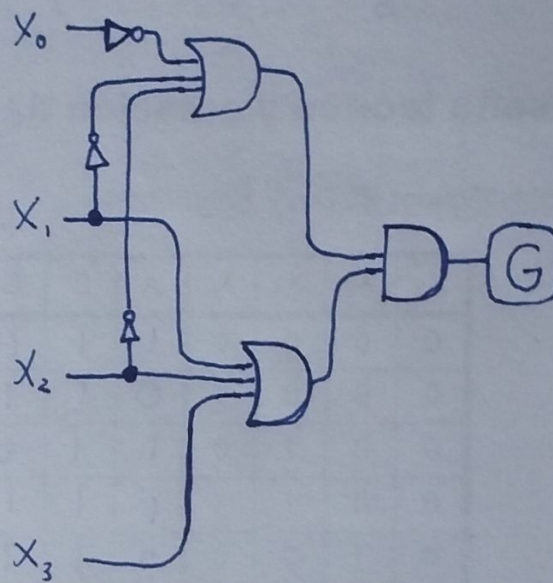
Función Lógica

$$\bar{G} = x_0 x_1 x_2 + \bar{x}_1 \bar{x}_2 \bar{x}_3$$

$$G = \bar{x}_1 \bar{x}_2 \bar{x}_3 + x_0 x_1 x_2 = (\bar{x}_1 \bar{x}_2 \bar{x}_3)(x_0 x_1 x_2) =$$

$$= (\bar{x}_1 + \bar{x}_2 + \bar{x}_3)(x_0 + x_1 + x_2) = (x_1 + x_2 + x_3)(\bar{x}_0 + \bar{x}_1 + \bar{x}_2)$$

Diagrama del Circuito



NAND

Segmento A

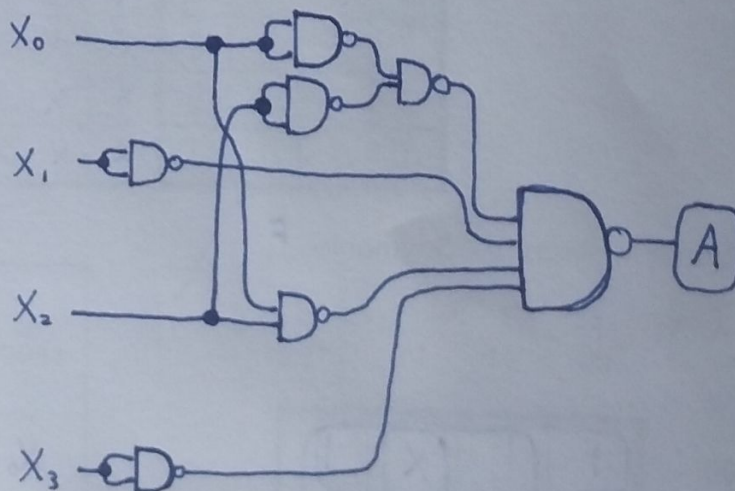
X_3, X_2	00	01	11	10
X_1, X_0				
00	1	0	X	1
01	0	1	X	1
11	1	1	X	X
10	1	1	X	X

Función Lógica

$$A = x_1 + x_3 + \bar{x}_0 \bar{x}_2 + x_0 x_2 =$$

$$= \bar{x}_1 \bar{x}_3 (\bar{x}_0 \bar{x}_2) (\bar{x}_0 x_2)$$

Diagrama del Circuito



NOR

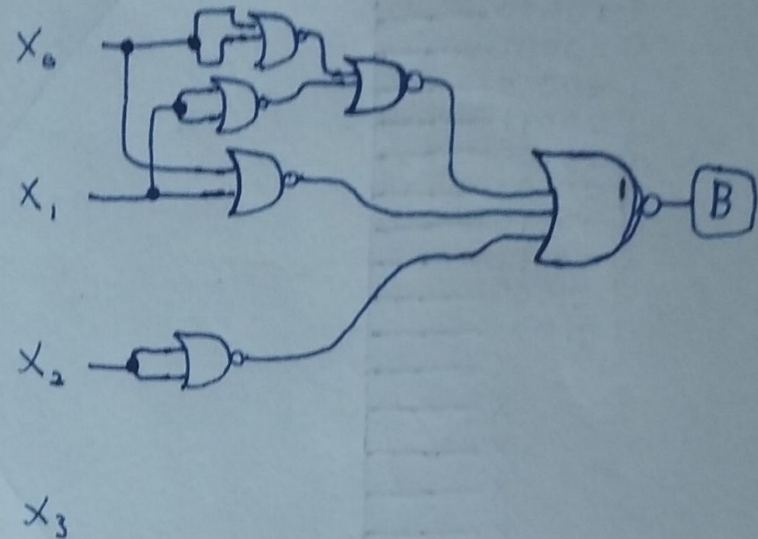
Segmento B

X_3, X_2	00	01	11	10
X_1, X_0				
00	1	1	X	1
01	1	0	X	1
11	1	1	X	X
10	1	0	X	X

Función Lógica

$$B = \overline{B} = \overline{x_2 + x_0 \overline{x_1} + x_0 x_1} = \overline{x_2} (\overline{x_0 \overline{x_1} + x_0 x_1}) = \overline{x_2} (\overline{x_0} \overline{\overline{x_1}} + \overline{x_0} \overline{x_1}) = \overline{x_2} (\overline{x_0} + \overline{x_1}) = \overline{x_2} + \overline{x_0} + \overline{x_1} = \overline{x_2} + \overline{x_0} + \overline{x_1} + \overline{x_0} + \overline{x_1}$$

Diagrama del Circuito



MUX4_1

Segmento C

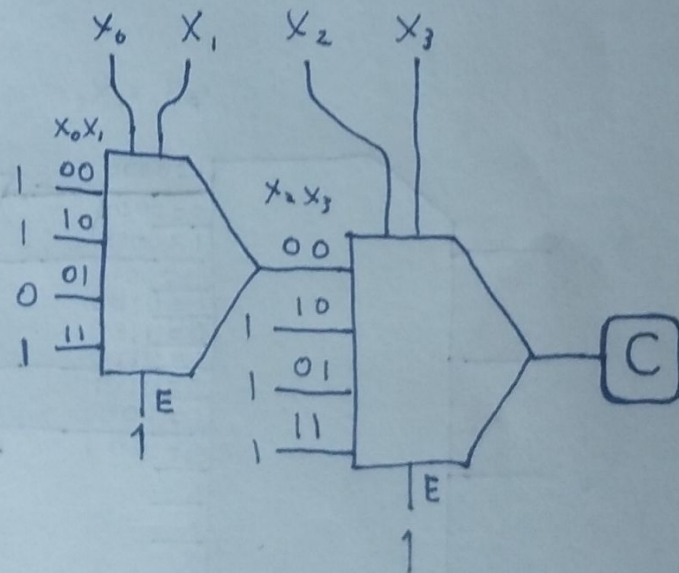
X_3, X_2	00	01	11	10
X_1, X_0				
00	1	1	X	1
01	1	1	X	1
11	1	1	X	X
10	0	1	X	X

Función Lógica

$$C(x_3, x_2, x_1, x_0) = (\overline{x_2} \overline{x_3}) C(x_0, x_1, 0, 0) + (\overline{x_2} x_3) C(x_0, x_1, 0, 1) + (x_2 \overline{x_3}) C(x_0, x_1, 1, 0) + (x_2 x_3) C(x_0, x_1, 1, 1) = (\overline{x_2} \overline{x_3}) C(x_0, x_1, 0, 0) + 1 + 1 + 1$$

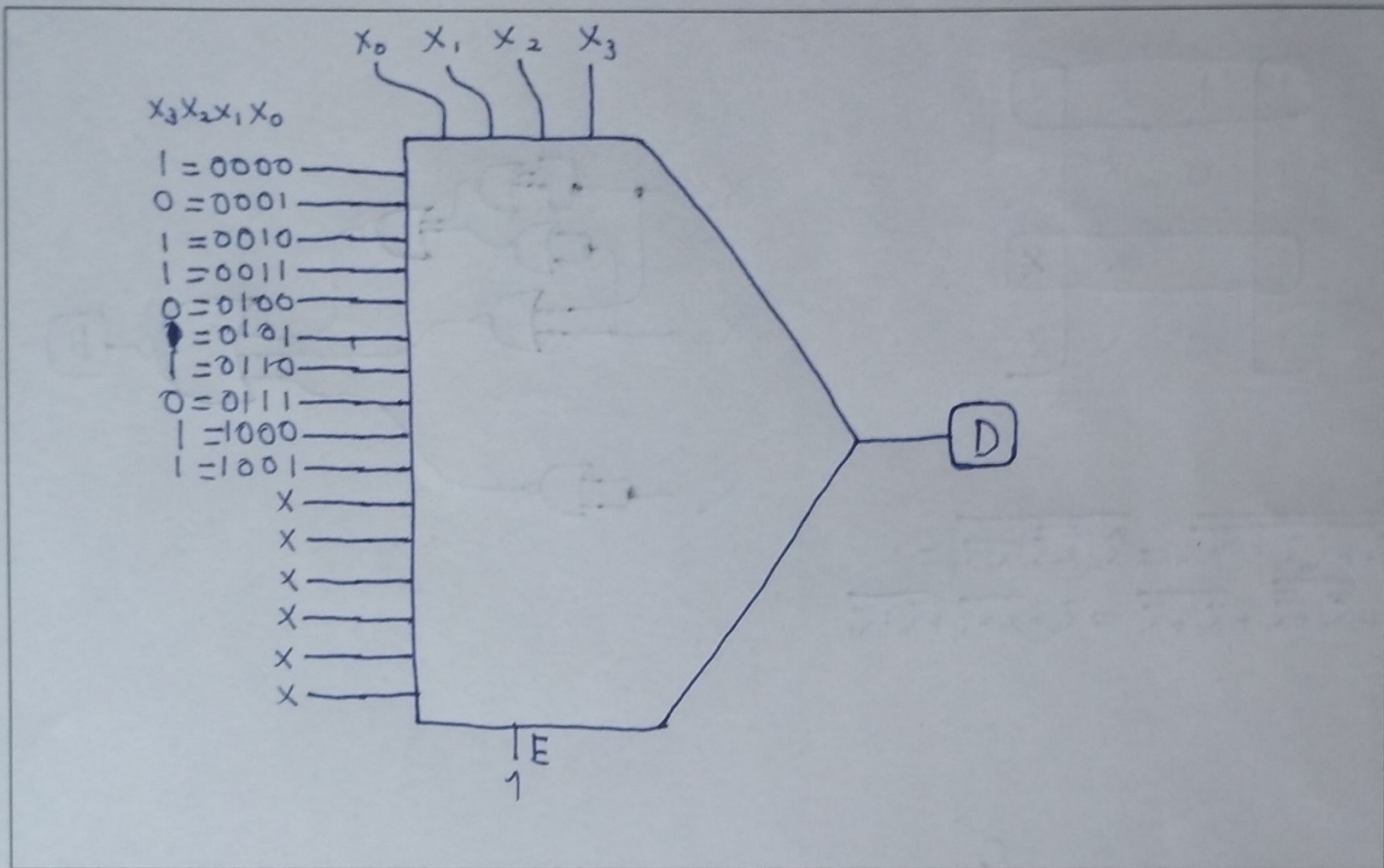
$$C(x_0, x_1, 0, 0) = (\overline{x_0} \overline{x_1}) C(0, 0, 0, 0) + (\overline{x_0} x_1) C(0, 0, 0, 1) + (x_0 \overline{x_1}) C(0, 1, 0, 0) + (x_0 x_1) C(1, 1, 0, 0) = 1 + 1 + 0 + 1$$

Diagrama del Circuito



MUX16_1 Segmento D

Diagrama del Circuito:



DEC Segmento E

Diagrama del Circuito:

